Alexandros Mylonakis

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Professional Summary

AI Engineer at NIKI Digital Engineering, onboarding onto engineering simulation and web-application projects. Integrated Master's graduate in Computer Science & Engineering (July 2026), with thesis research on gender bias in LLM embeddings and hands-on Python and NLP experience.

Education

University of Ioannina **July 2026**

Diploma in Computer Science & Engineering (Integrated Master) *Ioannina, Greece*

Experience

NIKI Digital Engineering **Sep. 2026 – Present**

AI Engineer *Ioannina, Greece (On-Site)*

- Onboarding onto client projects in engineering simulation and web applications, through code reading, shadowing and hands-on exploration of the project environments.
- Building working knowledge of MATLAB, TypeScript, React and JSON while collaborating with engineers across Niki and partner companies.

Ontovisual **May 2026 – June 2026**

Software Engineering Intern *On-Site / Ioannina, Greece*

- Architected an automated typesetting engine utilizing Shape Grammars and Guillotine Partitioning algorithms to deterministically map generative AI outputs into pixel-perfect, magazine-style UI pages.
- Designed structured prompt architectures to constrain LLM outputs into strictly formatted JSON payloads, seamlessly integrating with a custom client-side validation schema.
- Engineered a fault-tolerant deserialization pipeline with dynamic overflow and look-ahead buffer logic to translate serialized AI object states into continuous, multi-page component layouts.
- Developed adaptive typography guardrails to resolve Greek text-wrapping collisions, automatically truncating text in edit mode to guarantee zero element overlaps.

PanCar Rentals **2015 – 2022**

Fleet Operations & Client Service *Chania, Crete*

- Utilized strategic communication to de-escalate and resolve last-minute scheduling conflicts under strict time constraints, maintaining high customer satisfaction across 8 consecutive seasons.
- Demonstrated rapid adaptability in a fast-paced operational hub by onboarding new seasonal staff and adjusting fleet logistics to meet shifting daily demands.

University of Ioannina Radio Club **2017 – 2021**

IT Support & Web Administrator *Ioannina, Greece*

- Fostered effective teamwork by coordinating technical setups and logistics alongside cross-functional teams to deliver successful live music festivals and university seminars.
- Maintained and updated the station's digital infrastructure and website content, ensuring high uptime and accessibility for listeners.

Technical Projects

Master's Thesis: Measuring Bias in LLM2Vec Representations **Feb. 2025 – Feb. 2026**

Python, LLMs, NLP

- **Developed an automated diagnostic framework** to benchmark 6 public LLM2Vec-adapted bidirectional encoders against traditional Word2Vec and BERT baselines.
- **Built gender-bias axes** using L2-normalization and PCA; explained-variance analysis and comparison with single-pair (he–she) axes indicated that distributional methods are more stable for measuring bias in hidden states.
- **Compared model sizes and architectures**, observing that the structurally pruned Sheared-LLaMA-1.3B showed lower neutral-occupation bias than larger models, suggesting pruning may act as a denoising step.

Technical Skills

Languages: Greek (Native), English (C2 Certified - Michigan ECPE)

Programming Languages: Python, Java, C, JavaScript, TypeScript, MATLAB, HTML/CSS, SQL

Frameworks & Web: React, Redux, craft.js

AI & Data: Machine Learning, Deep Learning, NLP (LLMs), Data Mining, NumPy, Pandas, Scikit-learn, Databricks and Power BI (introductory)

Core Concepts: Computational Geometry, Space Partitioning, Layout Algorithms, Prompt Engineering

Developer Tools: Git, VS Code, Linux, Jupyter Notebooks, Claude Code, Antigravity